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4.4.7 Anther dehiscing direction

It shows the position of anther opening relative to the anther of the flower.

1.Introrse: Anther dehisces towards the center of the flower. Example: *Dianthus*.

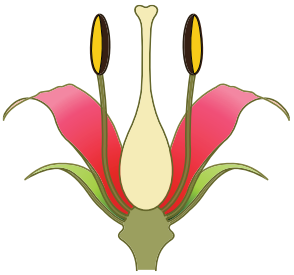


Figure 4.31: (a) Introrse



Figure 4.31: (b) Extrorse

2. Extrorse: Anther dehisces towards periphery of the flower. Example: *Argemone*.

4.5 Gynoecium

Gynoecium or pistil is the female reproductive part of the flower.

A pistil consists of an expanded basal portion called the ovary, an elongated section called a **style** and an apical structure that receives pollen called a **stigma**. Ovary with stipe is called **stipitate ovary.**

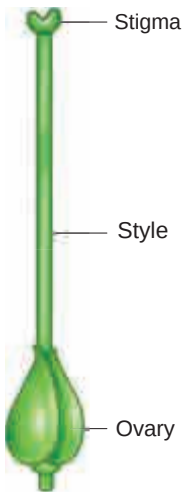


Figure 4.32: Pistil

Carpel: They are components of a gynoecium. Gynoecium is made of one or more carpels. Carpels may be distinct or connate.

4.5.1 Number of carpel

Unicarpellary (monocarpellary) Single carpel Example: Fabaceae	**Bicarpellary** Two carpels Example: Rubiaceae	**Tricarpellary** Three carpels Example: Cucurbitaceae	**Tetracarpellary** Four carpels Example: Lamiaceae.	**Multicarpellary** Many carpels Example: Nymphaeceae.

4.5.2 Fusion of carpels

It is an important systematic character. Apocarpous gynoecium is generally thought to be ancestral condition in Angiosperms.

Apocarpous A pistil contains two or more distinct carpels. Example: *Annona*.	**Syncarpous** A pistil contains two or more carpels which are connate. Example: Citrus, tomato.

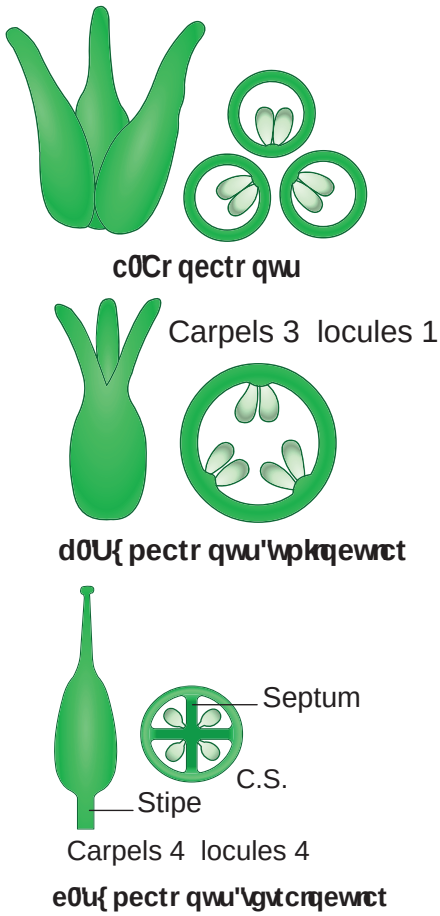


Figure 4.33: Fusion of carpels